Magnetically Steered Cardiac Ablation

ECGR 3120: Final Technical Report

Dashia Boyarskiy
Instructor: Dr. Joy
UNC Charlotte
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Method: All five milestones implemented against the provided instructor starter kit; 29/29 supplied acceptance tests passed. Numeric results are *computed*, not measured. The term "lesion volume" is an electrical power-density threshold proxy, not a thermal or clinical prediction.

Executive summary and verification scope

The system couples a two-coil magnetic navigation field to catheter-tip torque, a supplied geometric seating-depth rule, and tissue electrical conduction. Coil resistance and mounting force quantify the field's engineering cost. We completed the 36 previously empty function bodies in the instructor-provided source interfaces; the supplied 'check.py' suite reports **29 of 29 tests passing**. The original provided constants and supplied solver/bookkeeping routines were retained.

Baseline: coil radius and separation 0.350 m; 155 turns and 200 A per coil (NI = 31,000 A-turns). The center field is 79.642 mT. Electrical loss is 1.451 kW per coil and intercoil attraction is 690.3 N. At an illustrative design comparison of 40/80/160 mT, stronger fields improve steering and increase the power-density volume proxy, but sharply increase coil power and force.

Validation: the supplied tests check on-axis Biot–Savart predictions, force/torque balance, Ohmic and mechanical scaling, finite-volume resistance and material interface conditions, plus end-to-end integration. The additional M4 grid comparison is not a statistical confidence interval.

Scope: Every result below is a numerical computation using specified or explicitly documented model assumptions; no experimental hardware or tissue measurements were made.

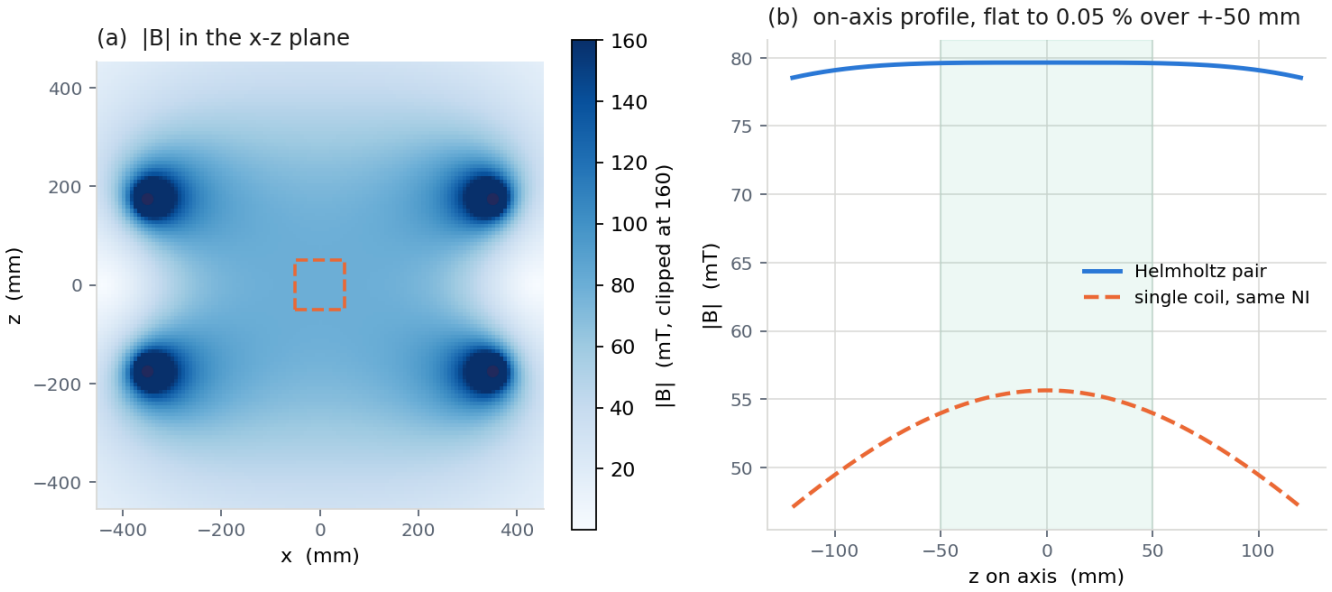
Important distinction: The previous standalone reports used assumptions before the starter kit was available. This final document uses its supplied baseline, including the 25 V electrode voltage, 40 mm domain radius, 1.2×10^6 W/m^3 threshold and seating-depth formula.

Milestone 1 — Magnetostatic navigation field

The magnetic field from each circular current loop is computed by the vectorized midpoint Biot–Savart sum

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 N I}{4\pi} \sum_s \frac{\Delta \mathbf{l}_s \times (\mathbf{r} - \mathbf{r}_s)}{|\mathbf{r} - \mathbf{r}_s|^3}.$$

The closed-form on-axis reference is $B_z(z) = \mu_0 N I a^2/[2(a^2+z^2)^{3/2}]$; the Helmholtz-pair field is the sum from planes $z = \pm a/2$. At coil radius/separation $a = 0.350$ m, 155 turns and 200 A, the computed central field is **79.642 mT**, versus **79.641 mT** analytically, within approximately 0.001%. The computed axial field-span variation over ±50 mm is 0.0469%; the supplied finite cube sample gives approximately 0.249%.



Increasing segment resolution reduces the on-axis numerical discrepancy (24 segments: 1.244%, 768 segments: 0.00121% for the single-coil test).

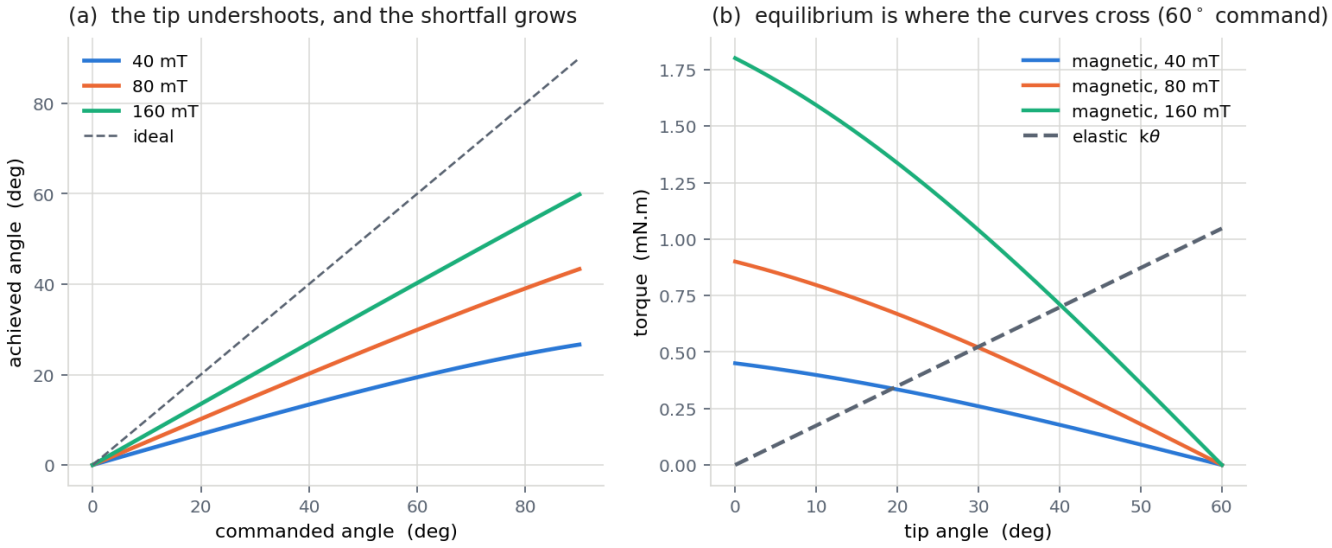
Validation: all six M1 acceptance tests passed; thin-filament coils neglect winding cross-section and edge fields near wire.

Milestone 2 — Magnet torque and catheter steering

The tip contains a cylindrical NdFeB slug with effective moment $m \approx 0.0130$ A·m^2. An equivalent current loop verifies the Lorentz-force and torque integrals against $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$. The shaft is approximated as a linear torsional spring, yielding equilibrium

$$k\theta_t = mB\sin(\theta_c - \theta_t), \qquad k = 1.0 \times 10^{-3}\ \text{N m/rad}.$$

With a 60° commanded angle, computed achieved angles are approximately 19.4° (40 mT), 29.9° (80 mT), and 40.3° (160 mT). The loop-integration torque and dipole torque agree; a closed loop experiences negligible net force in a uniform field.



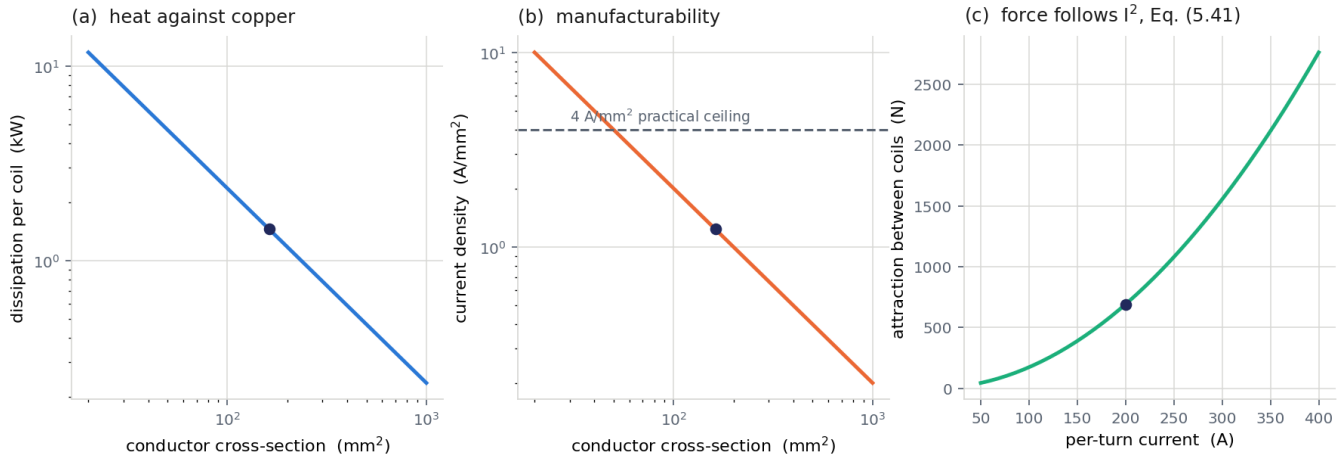
Validation: six M2 tests passed. The static spring model omits friction, bending modes, contact force and time-varying motion.

Milestone 3 — Electrical budget, heating and coil attraction

The baseline winding uses $N = 155$, $I = 200$ A and copper conductor area $A = 1.62 \times 10^{-4}$ m^2; length per coil is $L = N(2\pi a) = 340.9$ m. Resistance and dissipation are

$$R = \frac{L}{\sigma_{Cu} A} = 0.03628\ \Omega, \qquad P = I^2 R = 1.451\ \text{kW per coil.}$$

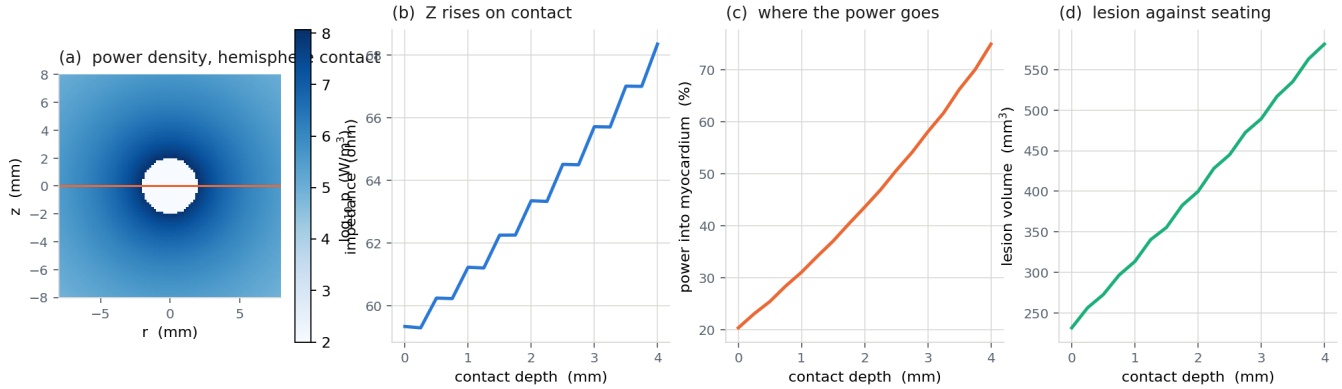
Computed current density is 1.23 A/mm^2. An integrated Biot–Savart/Lorentz calculation gives **690.3 N attraction** between the coils. The simple natural-convection relation $\Delta T = P/(hA_{exposed})$ predicts approximately **260 K rise** at baseline $h = 4$ W/(m^2 K); an uncooled assembly is therefore not a viable hardware design under these assumptions.



Validation: six M3 tests passed, including I^2 power and force scaling. The heating estimate does not model radiation, conductor temperature dependence, or forced cooling.

Milestone 4 — Electrical conduction in blood and myocardium

An axisymmetric finite-volume solver computes $\nabla \cdot (\sigma \nabla V) = 0$ for myocardium $\sigma = 0.54$ S/m and blood $\sigma = 0.70$ S/m, with a 2 mm spherical electrode and a 40 mm grounded domain. The electrode is held at 25 V. The electric power-density contour $p = \sigma |\mathbf{E}|^2$ is thresholded at 1.2×10^6 W/m^3 to define a *proxy* for lesion volume; actual thermal damage is not solved.



At a centered interface, the analytic resistance is 60.97 Ω and numerical resistance converges as follows:

Cells per electrode radius	Computed impedance	Relative error
6	65.52 Ω	7.46%
12	63.35 Ω	3.91%
24	62.09 Ω	1.84%

The reported 12-cell insertion sweep gives approximately 59–68 Ω, 20–75% myocardium power share and 232–581 mm^3 threshold-volume proxy. Centered split is 43.5%, in accord with $0.54/(0.54 + 0.70)$. Seven supplied M4 tests passed.

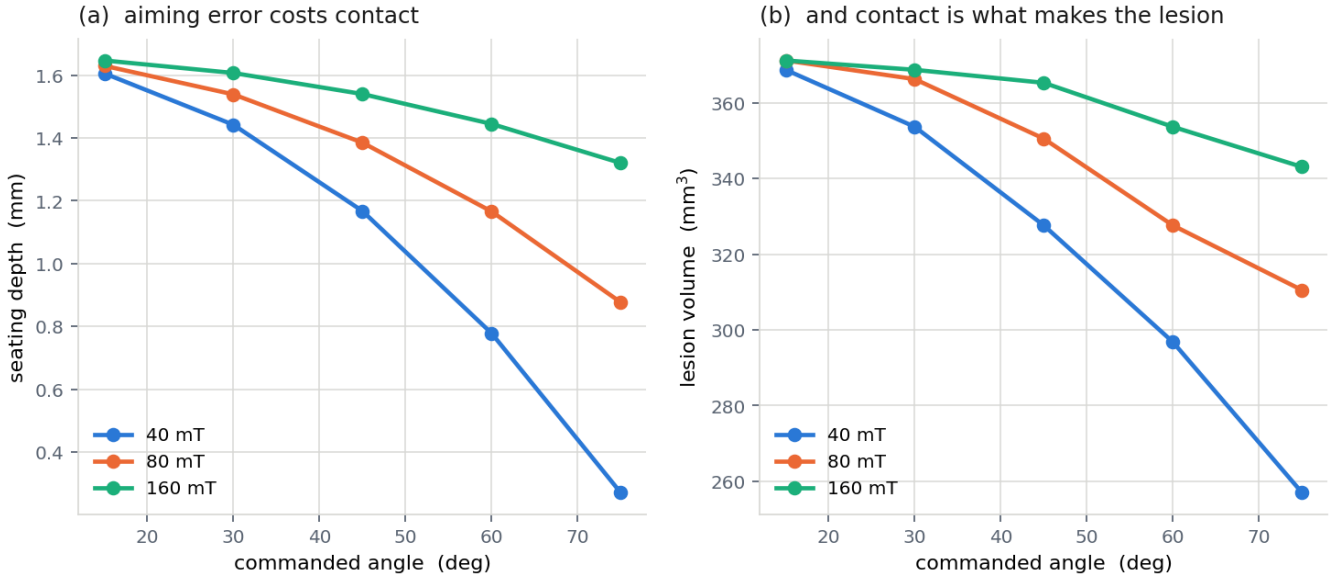
Limitations: staircase geometry, isotropic tissue, fixed voltage/boundary, and absence of bioheat, perfusion and thermal dose.

Milestone 5 — Coupled system response

The supplied geometry implements the angle-error to insertion-depth relation

$$d = \mathrm{clip}_{[0,2a]}[s\cos e - a(1-\cos e)]\,, \qquad a = 2 \text{ mm},\ s = 1.66 \text{ mm}.$$

For each commanded angle, the M2 model produces achieved tip angle and error; this relation determines electrode insertion depth for the M4 conduction solver. The official driver evaluates commands $15°, 30°, 45°, 60°, 75°$ at fields 40/80/160 mT.



At the most difficult tested 75° command, 12-cell computed proxy volume is 257.0 mm^3 (40 mT), 310.5 mm^3 (80 mT), and 343.1 mm^3 (160 mT). Increasing field reduces angular undershoot, improves normal seating and increases modeled power-density volume; the benefit must be weighed against rapidly increasing coil dissipation and attraction. All four official M5 tests passed.

The seating law is a starter-kit geometric assumption and is not a validated contact-force or patient-specific model.

Numerical sensitivity and design comparison

The limiting 75° case was recomputed on 6/12/24 cells per electrode radius, with the original supplied geometry and independent M4 solves for each evaluated field.

B (mT)	Depth (mm)	6 cells (mm^3)	12 cells (mm^3)	24 cells (mm^3)
40	0.2715	241.79	257.04	264.26
80	0.8773	301.01	310.54	315.30
160	1.3205	324.17	343.09	353.20

The observed 80 mT volume range, 301.01–315.30 mm^3, is a *finite-grid sensitivity range*, not a statistical confidence interval or bound on total error. It omits seating-law uncertainty, tissue anisotropy, threshold sensitivity and voltage uncertainty. Refinement of the centered-interface impedance from 6 to 12 to 24 cells gives 65.52, 63.35 and 62.09 Ω, indicating the expected decreasing staircase error.

Target field	Current per coil	Pair coil loss	Attraction
40 mT	$\sim$ 100.5 A	$\sim$ 0.73 kW	$\sim$ 174 N
80 mT	$\sim$ 201 A	$\sim$ 2.93 kW	$\sim$ 697 N
160 mT	$\sim$ 402 A	$\sim$ 11.7 kW	$\sim$ 2789 N

Power and force scale approximately with I^2 for fixed winding geometry. At 80 mT, the free-convection model predicts a temperature rise near 263 K; the cooling system must be redesigned.

Interpretation: A chosen minimum proxy must be explicitly justified as an illustrative screening objective; no threshold in this study is a clinical efficacy target.

Validation, interpretation, and limitations

Official acceptance: 6/6 tests passed for M1, 6/6 for M2, 6/6 for M3, 7/7 for M4 and 4/4 for M5, totalling **29/29**. The warm-up and M1–M5 scripts all ran and wrote numerical transcripts and PNG figures.

Independent checks: Biot–Savart field was compared with the analytical loop; magnetic loop torque was compared with $\mathbf{m} \times \mathbf{B}$; coil dissipation and attraction were checked against quadratic current scaling; tissue resistance was compared against the analytical spherical interface; M4 grid refinement and the M5 depth/field chain were tested.

Physical interpretation: The parallel-conductivity split gives $\sigma_m/(\sigma_m + \sigma_b)$ at hemispherical contact. As the electrode enters myocardium, it shifts electric power from blood to the wall but also changes resistance. Magnetic steering improves insertion for steep commanded directions at the price of expensive windings and support loads.

Critical limitations: (1) The finite-volume solution is steady, whereas RF ablation is typically around 500 kHz. The conduction-dominance condition $\sigma \gg \omega\epsilon$ must be checked against an explicitly sourced or assumed frequency-dependent permittivity before making a quantitative claim. (2) The power-density contour is not a biological lesion; absent are the bioheat equation, perfusion and tissue damage kinetics. (3) Conductivity is treated as homogeneous and isotropic within each tissue region, ignoring anisotropy, patients and temperature effects. (4) Winding turns are collapsed into ideal circular filament loops, not resolved multiconductor coils. (5) The steering-to-seating law is a geometric assumption, not a contact-force model. (6) No bench or clinical measurements validate the predictions.

Status of prior milestone reports: Their old illustrative numbers are retained as historical development work; the new official baseline governs this final report.

Verification and provenance: All numerical values are computed using the supplied starter-kit physical inputs and the implemented functions. They are not experimental measurements or independently certified device specifications.

Conditional engineering recommendation and next steps

For an explicitly *illustrative engineering screen*, choose a minimum of **290 mm**^{3} for the power-density threshold-volume proxy across the *five commanded angles* 15°–75° supplied by the instructor's integration driver. At the limiting 75° command, the observed refined-grid proxy ranges are 241.79–264.26 mm^3 at 40 mT, 301.01–315.30 mm^3 at 80 mT, and 324.17–353.20 mm^3 at 160 mT. The computed 12-cell command sweep decreases with increasing command angle, so 75° is the limiting tested case.

On this screening criterion, 80 mT is the lowest *evaluated* field passing the chosen proxy target. It requires roughly 201 A in each of two 155-turn coils; each dissipates about 1.46 kW, the pair about 2.93 kW, and attraction is around 697 N. The simple free-convection rise is around 263 K, making the existing uncooled arrangement unsuitable. Increased thermal management and mechanical design are necessary.

The 40 mT case is significantly less power-intensive but does not meet the illustrative target. The 160 mT case improves the proxy volume, but at approximately 11.7 kW pair power and 2.8 kN attraction, demonstrating diminishing engineering value for this particular screening requirement.

Not a medical or hardware recommendation: The 290 mm^3 screening target is chosen to compare numerical models; it is not specified in the handout or supported as a clinical damage threshold. An operational recommendation requires a validated contact model, actual cooling and support design, tissue properties, thermal-dose simulation, and patient safety assessment.

Next steps: (1) Verify the design criterion with Dr. Joy. (2) Explore stronger cooling and winding-pack thermal resistance. (3) Test contact-force and steering-depth sensitivity. (4) Move to a full transient bioheat/damage model before making any statement about clinical lesion size.

Appendix/source package: 'check.py' official output, per-milestone numerical logs, starter kit and implemented Python sources are included in the code archive for reproducibility.